

The Value of Volatility Timing (Question 9)

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Why Volatility Timing Matters

Prior literature:

- Volatility models often evaluated by statistical fit
- Standard models explain only a small fraction of variation in squared returns
- Low R^2 led many researchers to question the economic usefulness of volatility forecasts

Fleming, Kirby & Ostdiek (2001):

- Ask whether volatility timing has **economic value for asset allocation**
- Compare a *static* portfolio (fixed weights) with a *dynamic* portfolio (time-varying weights)
- Expected returns are difficult to estimate (Merton, 1980), so they are held constant and portfolio dynamics are driven by the covariance matrix

Economic intuition:

- Volatility forecasts determine how portfolio weights adjust over time
- The resulting performance improvement relative to a static portfolio is the economic value of volatility timing

Portfolio Allocation Framework

The investor's problem (mean-variance analysis):

$$\min_{\mathbf{w}_t} \mathbf{w}_t' \boldsymbol{\Sigma}_t \mathbf{w}_t \quad \text{s.t.} \quad \mathbf{w}_t' \boldsymbol{\mu} + (1 - \mathbf{w}_t' \mathbf{1}) R_f = \mu_p$$

Closed-form solution (in terms of excess returns):

$$\mathbf{w}_t = \frac{\mu_p \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}}{\boldsymbol{\mu}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}}$$

Interpretation:

- $\mathbf{w}_t$: time-varying portfolio weights on risky assets; $\boldsymbol{\Sigma}_t$: conditional covariance matrix; $\boldsymbol{\mu}$: expected returns (treated as *constant* at the daily horizon)
- All time variation in $\mathbf{w}_t$ is driven purely by changes in $\boldsymbol{\Sigma}_t$ — this *is* volatility timing
- Two strategies: (i) **minimum volatility** — fix target return μ_p , minimise variance; (ii) **maximum return** — fix target volatility σ_p , maximise expected return

Conditional Covariance Estimator

Conditional covariance estimator between assets i and j at time t (Foster & Nelson, 1996):

$$\hat{\sigma}_{ij,t} = \sum_{l=-t+1}^{T-t} \omega_{ij,t+l} (r_{i,t+l} - \mu_i) (r_{j,t+l} - \mu_j)$$

- $l < 0$: past observations; $l > 0$: future observations
- Connection to GARCH: substituting recursively into $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$:

$$\sigma_t^2 = \frac{\omega}{1-\beta} + \sum_{l=1}^{\infty} \underbrace{\alpha \beta^{l-1}}_{\omega_l} \underbrace{\varepsilon_{t-l}^2}_{(r_{t-l}-\mu)^2}$$

where $\varepsilon_{t-l} = r_{t-l} - \mu$. GARCH can be expressed as a weighted sum of past squared shocks with geometrically decaying weights $\omega_l = \alpha \beta^{l-1}$. The Foster–Nelson estimator uses a similar weighted-innovation structure and extends naturally to conditional covariances.

Exponential Decay Weights

Optimal weights (two-sided estimator):

$$\omega_{ij,t+l} = \frac{\alpha_{ij,t}}{2} e^{-\alpha_{ij,t}|l|}$$

- Optimal weights decline with $|l|$, where α is the decay rate:
 - Larger α emphasizes recent data and produces more responsive volatility estimates
 - Smaller α produces smoother and more persistent volatility estimates
- A common α is imposed across all (i, j) pairs to guarantee that the conditional covariance matrix is positive definite.
- This common α is estimated by minimizing the asymptotic MSE of the portfolio variance,

$$\sigma_{pt}^2 = \mathbf{w}_t' \boldsymbol{\Sigma}_t \mathbf{w}_t,$$

using the two-sided covariance estimator, yielding $\hat{\alpha} = 0.063$.

- The two-sided estimator is not implementable in real time; for portfolio construction, set $\omega_{ij,t+l} = 0$ for $l > 0$ and double the remaining weights.

Measuring Economic Value

Average realised utility under quadratic preferences:

$$\bar{U}(\cdot) = W_0 \sum_{t=0}^{T-1} \left(R_{p,t+1} - \frac{\gamma}{2(1+\gamma)} R_{p,t+1}^2 \right)$$

where γ is constant relative risk aversion and $R_{p,t+1} = R_f + \mathbf{w}'_t \mathbf{r}_{t+1}$.

Estimating the performance fee Δ : find Δ such that

$$\sum_{t=0}^{T-1} \left[(R_{d,t+1} - \Delta) - \frac{\gamma}{2(1+\gamma)} (R_{d,t+1} - \Delta)^2 \right] = \sum_{t=0}^{T-1} \left[R_{s,t+1} - \frac{\gamma}{2(1+\gamma)} R_{s,t+1}^2 \right]$$

with static portfolio return $R_{s,t+1}$ and dynamic portfolio return $R_{d,t+1}$

Interpretation:

- $\Delta > 0$ means the investor would pay to switch from the static to the dynamic strategy, so volatility timing has positive economic value
- Reported as annualised basis points for $\gamma \in \{1, 10\}$, quantifying how risk aversion shapes the benefit

Data and Preliminary Analysis

Data: S&P 500, T-Bond, and Gold futures, 1983–1997 (3,763 daily returns)

Key facts:

- Futures returns = excess returns $\Rightarrow$ cash position implicit
- Stocks: $\mu = 10.8\%$, $\sigma = 16.2\%$; Bonds: $\mu = 6.5\%$, $\sigma = 10.5\%$; Gold: $\mu = -7.8\%$, $\sigma = 14.7\%$
- Stock-bond correlation: $+0.40$; Stock-gold: -0.10 ; Bond-gold: -0.16

Conditional volatility (Foster–Nelson estimates):

- **Strong persistence:** Autocorrelations of vol. change decay slowly
 - S&P 500: $\rho_1 = 0.96$, $\rho_{10} = 0.50$
 - T-Bond: $\rho_1 = 0.92$, $\rho_{10} = 0.43$
 - Gold: $\rho_1 = 0.93$, $\rho_{10} = 0.41$
- Volatility changes are positively correlated across markets ($\rho(\sigma_t) \approx 0.2\text{--}0.5$)
- Correlations vary over time (stock-gold turns more negative post-1987)

Implication: Volatility is **persistent**, **predictable** $\Rightarrow$ volatility timing is feasible.

Baseline Results: Dynamic Portfolio Performance

Setup: Expected returns = full-sample means (negligible estimation risk)

Portfolio rules (daily rebalancing):

- *Minimum volatility:* Target 10% expected return
- *Maximum return:* Target 12% volatility

Ex post performance (full sample):

Strategy	μ (ann.)	σ (ann.)	Sharpe
Min Vol	9.75%	10.65%	0.916
Max Return	12.38%	13.57%	0.912
Static efficient portfolio (ex post)			0.86

Key finding:

- Ex post efficient frontier for static portfolios $\Rightarrow$ max Sharpe = 0.86
- **Both dynamic strategies plot above the static efficient frontier**

Statistical Significance: Permutation Test

Problem: Dynamic strategies use the same data for estimation and evaluation $\Rightarrow$ risk of spurious results.

Solution: Permutation test

- Randomly shuffle returns (break link between weights and returns)
- Apply *actual* weights to shuffled returns
- Repeat 10,000 times

Logic: If volatility timing has no value, the observed pairing of weights and returns is just one random draw among many.

Results:

Strategy	% trials with higher Sharpe
Min Vol	9.2%
Max Return	0.4%

- **Min Vol:** No shuffled trial achieved volatility as low as actual
- **Max Return:** Only 0.4% beat the actual Sharpe

Conclusion: Volatility-timing strategies perform far better than random chance.

Subsample Analysis and Crash Robustness

Why this analysis?

- To ensure results are not driven by a specific period (e.g., 1990s bull market)
- To test the impact of the October 1987 crash on performance

Sharpe ratios by 3-year subperiod:

Period	Min Vol	Max Return
1983–1985	0.905	0.806
1986–1988	0.712	0.520
1989–1991	0.879	0.682
1992–1994	0.555	0.522
1995–1997	1.892	2.060

Robustness to the 1987 crash:

Sample	Min Vol	SR	Max Ret	SR
Full sample	0.916		0.912	
Exclud. Oct 19–30, 1987	0.878		0.883	
Exclud. 1986–1988	1.036		1.019	

Observations:

- Strong cross-period variability: Sharpe ratios range from 0.52 to 2.06
- The 15-year average performance is the most reliable measure

Interpretation:

- Excluding the two crash weeks leaves Sharpe ratios almost unchanged
- Excluding the 1986–1988 period (crash period) even slightly improves performance

Conclusion: results are not driven by the 1987 extreme event

How Much Does Estimation Risk Matter?

Problem: In baseline, investors know true mean returns. In reality, they must estimate them from limited past data — introducing estimation risk that must be quantified.

Bootstrap setup: Draw k returns (with replacement), estimate mean returns, compute portfolio weights, apply them to actual returns, and repeat 10,000 times for each k .

Key findings (for $k = 4,000$, comparable to sample size):

	Min Vol Strategy			Max Return Strategy		
	μ	σ	SR	μ	σ	SR
Performance	8.73	10.46	0.840	11.58	13.57	0.853
Fee Δ (bp)	$\Delta_1 = 105, \Delta_{10} = 131$			$\Delta_1 = 80, \Delta_{10} = 80$		

Interpretation:

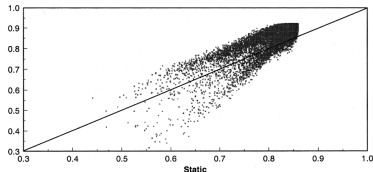
- In this case, an investor with $\gamma = 1$ would pay **80–105 bp annually** to eliminate estimation risk, which is not negligible
- But as k increases, performance approaches the "no estimation risk" case
- For small k , $\Delta_1 > \Delta_{10}$: low risk averse investors suffer more because both μ and σ deteriorate

Empirical Evidence: Volatility Timing

Panel A: Minimum Volatility Strategy

Panel A: Minimum Volatility Strategies

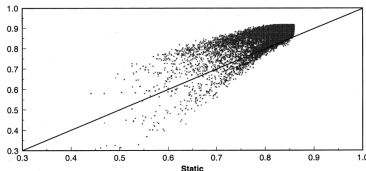
Dynamic



Panel B: Maximum Return Strategy

Panel B: Maximum Return Strategies

Dynamic



Key Results

- Each point represents a bootstrap simulation that compares Sharpe ratios of static and dynamic portfolios.
- The 45-degree line indicates equal performance.
- Dynamic portfolios outperform static portfolios in:
 - **84%** of simulations (minimum volatility strategy)
 - **92%** of simulations (maximum return strategy)
- Most observations lie above the line, suggesting that dynamic strategies achieve higher Sharpe ratios.

Economic Value of Volatility Timing

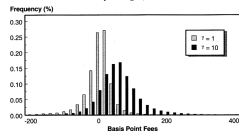
Interpretation

- Figure 3 quantifies the **economic value** of volatility timing.
- It reports the **performance fee** that investors would be willing to pay to use dynamic strategies instead of static portfolios.
- The fee is defined as the amount that makes investors **indifferent between the two strategies**.

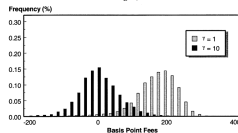
Key Findings

- The distribution of performance fees is mostly positive.
- Depending on their risk aversion, investors would pay on average **-2-170 bps per year**.
- This indicates that volatility timing often provides **substantial economic value**.

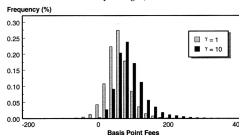
Panel A: Minimum Volatility Strategies, $\alpha = 0.063$



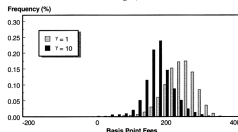
Panel B: Maximum Return Strategies, $\alpha = 0.063$



Panel C: Minimum Volatility Strategies, $\alpha = 0.005$



Panel D: Maximum Return Strategies, $\alpha = 0.005$



Distribution of performance fees across bootstrap simulations ($k=4,000$)

Sensitivity Analysis

Experimental Procedure

- The authors vary the decay rate across a wide range:

$$\alpha \in [0.001, 0.1]$$

- For each value of α , they repeat the empirical analysis:
 - ① Estimate the conditional covariance matrix using the new decay rate.
 - ② Construct dynamic portfolios based on volatility forecasts.
 - ③ Run bootstrap simulations to evaluate portfolio performance.

Key Findings

- Volatility timing performs **even better** when the decay rate is smaller.
- Smoother volatility estimates lead to **higher Sharpe ratios and greater economic value**.
- The main results remain **robust across different parameter choices**.

Conclusions

Main Findings

- Volatility is predictable and can be used to improve portfolio allocation.
- Dynamic volatility timing strategies outperform static portfolios.
- The Sharpe ratio increases when investors adjust portfolio weights based on time-varying volatility.

Economic Significance

- Investors would overall be willing to pay to implement volatility timing strategies.
- This suggests that volatility forecasts have **substantial economic value**.

Implication

- Even if volatility models are imperfect statistically, they can still provide **valuable guidance for dynamic asset allocation**.

References

Fleming, J., Kirby, C. and Ostdiek, B. (2001) 'The economic value of volatility timing', *Journal of Finance*, 56(1), pp. 329-352.